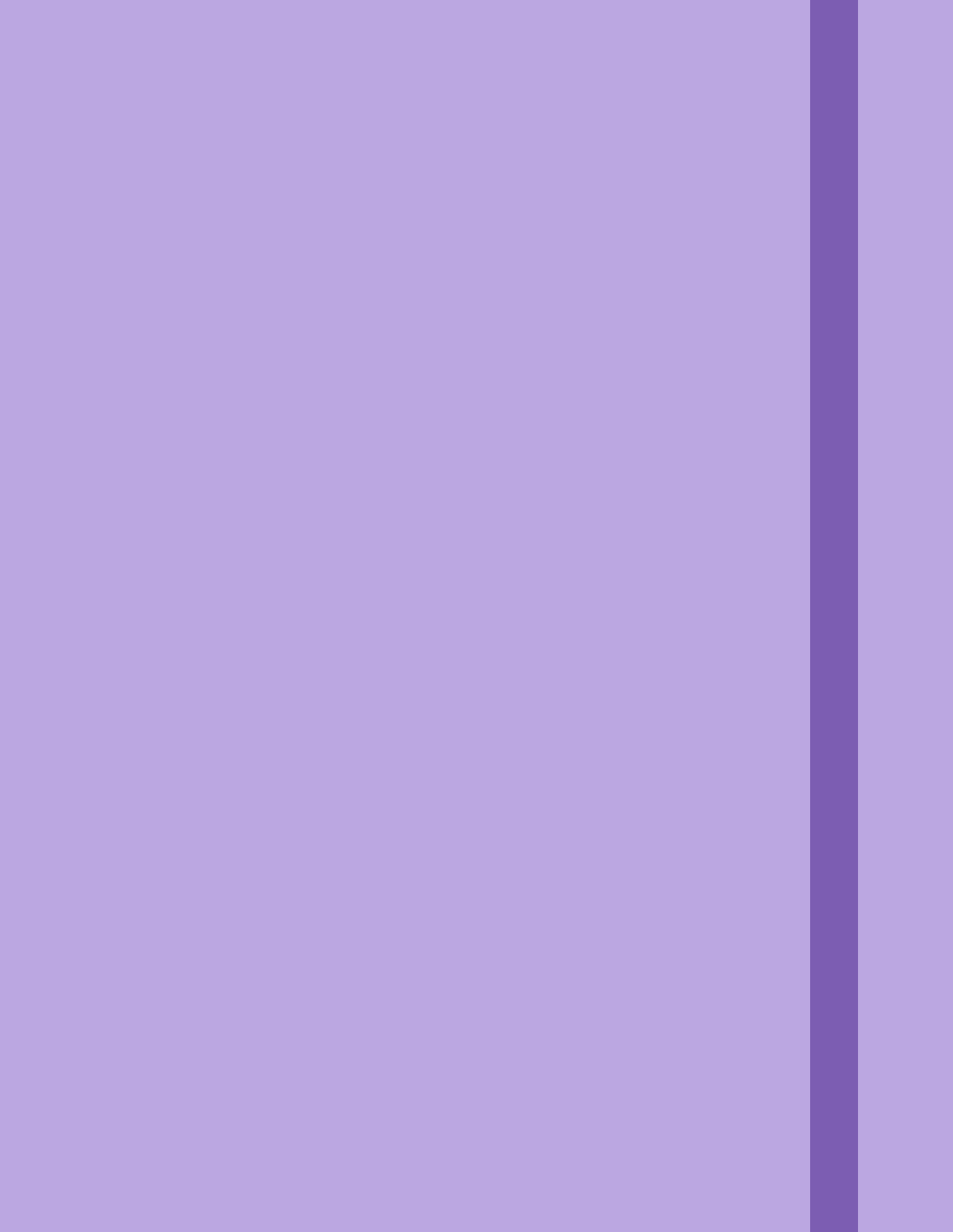
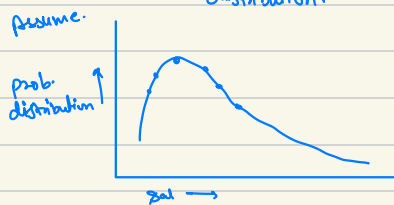




Sampling Distribution.

Sampling distribution is a probability distribution that describes the statistical properties of a sample statistics (such as a sample mean or sample proportion) computed from multiple independent samples of the same size from a population.

eg.) India people salary $\rightarrow$ population
25k, 100k, 0, 50k etc.
 $\downarrow$
distribution.



Sample = 50 people sal. $\rightarrow$ mean₁

Sample 2 = 50 people sal. $\rightarrow$ mean₂

$\vdots$

Sample 100 = 50 people sal. $\rightarrow$ mean₁₀₀

$\downarrow$
Sampling distribution of
sample mean.

Why sampling distribution is important?

In ML it allows us to estimate the variability of a sample statistic, which is useful for making inferences about population.

By analyzing the properties of the sampling distribution, we can compute confidence interval, perform hypothesis testing, and make prediction about the population on the sample data.

Central Limit Theorem.

The CLT states that distribution of the sample means of a large no. of independent & identically distributed random variables will approach a normal distribution, regardless of underlying distribution of variables.

The conditions required for CLT to hold are:

1. The sample size is large enough, typically greater than or equal to 30.
2. The sample is drawn from a finite population or an infinite population with a finite variance.
3. The random variables in the sample are independent & identically distributed.

The CLT is imp. in stats. & ML because it allows us to make probabilistic inferences about a population based on sample of data. For. g., we can use the CLT to construct confidence intervals, perform hypothesis tests, & make prediction about the population mean based on the sample data.

The CLT also provides a theoretical justification for many commonly used statistics techniques such as t-tests, ANOVA, and linear regression.

$$\begin{array}{ccc} \text{Any} & & \text{Normal.} \\ \mu, \sigma^2 & \longrightarrow & \mu, \frac{\sigma^2}{n} \end{array}$$

Confidence Interval.

Revision

Population	vs.	Sample.
↓		↓
every one included.		Random Selection

Parameter vs. Estimate/Statistic

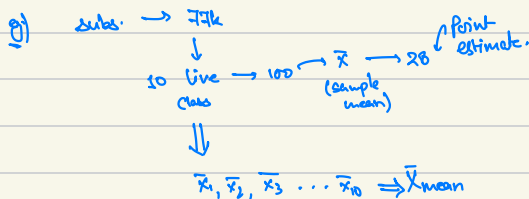
Population { mean, std, etc }
↓
parameter

Sample { mean, std, etc }
↓
statistic

Inferential Statistic.
↓ sample
↓ prediction.

Point Estimate.

A point estimate is a single value calculated from a sample that serves as the best guess or appr. for an unknown population parameter, such as mean or std. Point estimates are often used in stats. when we want to make inferences about a population based on sample.



Confidence Estimate

eg) If Dhoni makes 25 runs.
if exactly predicted $\xrightarrow{\text{get}}$ Rs. 10 Cr.
if $30 \pm$ $\rightarrow$ Rs. 15 lakh
if $20 \pm$ $\rightarrow$ Rs. 50 lak.

Point estimates are not reliable.

$\downarrow$
range

$\text{range} \rightarrow \text{CI}$

Confidence Interval : in simple words, is a range of values within which we expect a particular population parameter, like mean, to fall. It's a way to express the uncertainty around an estimate obtained from a sample of data.

Confidence level : usually expressed as a percentage like 95%, indicates how sure we are that the true value lies within the interval.

$$\text{C.I} = \text{Point Estimate} + \text{Margin of Errors.}$$

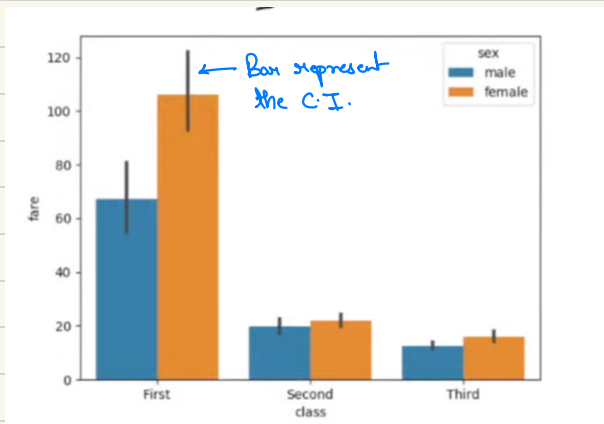
Ways to calc. C.I :

1) Z procedure
Applied when
we have std

2) t-procedure
Applied when we
don't have std
(standard deviation)

C.I is created for parameters and not statistics. Statistics helps us get the C.I for a parameter.

Eg. of C.I. usage :



Bar Plot

Confidence Interval (Sigma known) / Z-procedure :

Assumptions :

1. Random Sampling $\rightarrow$ Data collected must be random.
2. Known population standard deviation.
3. Normal distribution $\rightarrow$ The Z-procedure assumes that underlying population is normally distributed. However, if the population distribution is not normal, the Central Limit Theorem is applied if the sample size is large ($n \geq 30$).

Formula for calculating C.I. using Z-procedure :

$$\text{Point Estimate} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$(\bar{x})$

$Z = ?$

$1-\alpha =$ Confidence level. $\sigma =$ std $n =$ sample size.

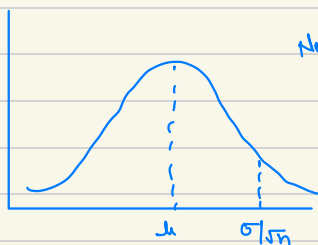
Intuition for above formula:

Point Estimate ($\bar{x}$) $\rightarrow$ CLT

10 live class $\rightarrow$ 50 students in each class.

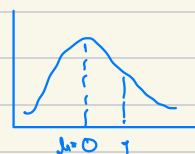
$\downarrow$ give us.

$[\bar{x}_1 + \bar{x}_2 + \dots + \bar{x}_{10}] \rightarrow$ sampling distribution of sample mean.

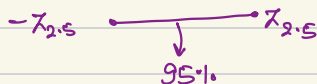
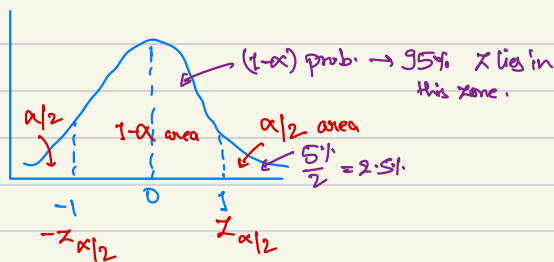


Normally distributed. $\xrightarrow{\text{to}}$ Standard Normal Variable.

$$\left(Z = \frac{\bar{x} - \mu}{\sigma} \right)$$



Now we need range with [confidence level $(1-\alpha) = 95\%$]



$$\text{probab. } (-Z_{\alpha/2} < Z < Z_{\alpha/2}) = 1-\alpha$$

$$\text{prob. } \left(-Z_{\alpha/2} < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2} \right) = 1-\alpha$$

$$\text{prob. } \left(-Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{x} - \mu < Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1-\alpha$$

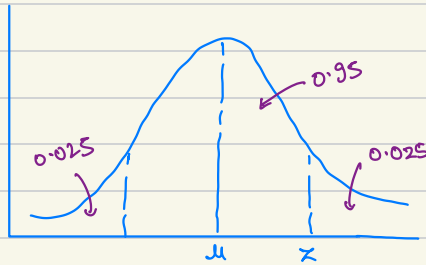
$$\text{prob.} \left(-\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < -\mu < -\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha$$

$$\text{prob.} \left(\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \boxed{\mu} < \bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha$$

↪ fixed

$\bar{X}$ = sample mean.

for calculating z :



$$(1 - 0.025) = 0.9750$$

z table value of 0.9750 = 1.96.

$$\mu = \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}$$

we get → range
↵ is called
Confidence
Interval (C.I)

Interpreting Confidence Interval :

i. Confidence Level →

g) 95% → (18 to 42) age group.

From 77k subs. → 100 times
random sample (50)

My age 1
⋮
My age 100 } 95% times
 } the age will
 } fall (18 to 42)

2. Interval range. $\rightarrow$ Width of Confidence Interval.

• If Confidence level $\uparrow \rightarrow$ the C.I $\uparrow$.
↳ vice-versa.

Factors affecting Margin of Error :

1. Confidence level $(1-\alpha)$

2. Sample size (n)

3. Population std. (σ)

C.I = Point Estimate + Margin of Error.

$$C.I = \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Annotations:

- $\bar{X}$: Sample mean
- $Z_{\alpha/2}$: Critical value.
- σ : pop. std
- $\sqrt{n}$: Sample size

Problem with $\bar{X}$ -procedure : We generally do not have std of population

Confidence Interval (Sigma not known) | t-procedure.

Using the t procedure :

Assumptions :

1. Random Sampling.
2. Sample standard deviation.
3. Appx. normal distribution.
4. Independent Observations.

t-procedure :

$$\bar{X}\text{-procedure} \Rightarrow C.I = \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \leftarrow \text{population std.}$$

$$t\text{-procedure} \Rightarrow C.I = \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{n}} \quad \leftarrow \text{sample std}$$

$$C.I = \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

It becomes student's T-distribution
It seems as normal dis. but is not exactly normal dis.
It's uncertain

- T-distribution is theoretical distribution, it's not present in nature.
T-distribution depends upon single parameter : degree of freedom (n-1)

Tail is fatter, more outliers.

Now we use t-table.